

Generative AI and the future of education

A recent UNESCO research paper titled *Generative AI and the future of education* (<https://unesdoc.unesco.org/ark:/48223/pf0000385877.locale=en>) is a must-read for anyone in the field of education. While it does recognise the potential of AI chatbots to mount language barriers, it clearly warns that generative AI is quite different from the AI tech that fuels search engines. Here, the AI chatbot gives an independent response, learnt from disparate sources, but not attributable to any single person. This is a cause of concern. Plus, when students start getting answers (right or wrong) from a chatbot, how will it affect human relationships? Will it undermine the role of teachers in the education ecosystem? How can institutions and nations enforce proper regulations to prevent malpractices? These and more such important issues, related to the use of generative AI in education, are discussed in the report.

“The speed at which generative AI technologies are being integrated into education systems in the absence of checks, rules or regulations, is astonishing... In fact, AI utilities often required no validation at all. They have been ‘dropped’ into the public sphere without discussion or review. I can think of few other technologies that are rolled out to children and young people around the world just weeks after their development... The Internet and mobile phones were not immediately welcomed into schools and for use with children upon their invention. We discovered productive ways to integrate them, but it was not an overnight process. Education, given its function to protect as well as facilitate development and learning, has a special obligation to be finely attuned to the risks of AI—both the known risks and those only just coming into view. But too often we are ignoring the risks,” writes Stefania Giannini, Assistant Director-General for Education, UNESCO, in her July 2023 paper.

els from Microsoft Azure OpenAI Service. AI4Bharat is an open source language AI centre based at the Indian Institute of Technology Madras (IIT-M). “We saw Jugalbandi as a kind of *chatbot plus plus* because it is like a personalised agent. It understands your exact problem in your language and then tries to deliver the right information reliably and cheaply, even if that exists in some other language in a database somewhere,” remarked Abhigyan Raman, a project officer at AI4Bharat, in a press release from Microsoft.

During the trials, Microsoft found rural Indians using the chatbot for various purposes like discovering scholarships, applying for pensions, figuring out the status of someone’s government assistance payments, and so on.

Call from your business partner? Are you sure?

While such developments are heartening, the number of scams happen-

ing with inexpensive AI tools as an ally is alarming!

A few months ago, an elderly couple in the US got a call from their grandson, who seemed grief-stricken. He said he was in jail and needed to be bailed out with a huge sum of money. The couple frantically set about withdrawing money from their bank accounts. Seeing this, a bank manager took them aside and enquired what the problem was. He smelled something fishy and asked them to call their grandson back on his personal number. When they called him, they found out that he was safe and sound! The earlier call was from an imposter—and obviously from an unknown number, as he claimed he was in jail and calling from the police station! What was more, the imposter sounded exactly like the grandson they had been speaking to for decades, and the couple hardly found anything amiss. Who was his partner in crime? A cheap AI tool which, when fed a few samples of

audio recordings by a person, can replicate their voice, accent, tone and even word usages quite precisely! You can type in what you want it to say, and it will say it exactly like the person you wish to mimic.

These kinds of scams using deepfake voices are not new. Way back in 2020, a bank manager in Hong Kong authorised a transfer of HK\$35 million, duped by a phone call from someone who sounded exactly like one of the company’s directors! An energy company in UK lost a huge sum when an employee transferred money to an account, purportedly of a Hungarian supplier, following orders from his boss (a voice clone!) over the phone.

At that point of time, it took time and resources to create a deepfake, but today, voice synthesising AI tools like that from ElevenLabs require just a 30-second audio clip to replicate a voice. The scammer just needs to grab a small audio clipping of yours—could be from a video posted on social media, a recorded lecture, or a conversation with some customer care service (apparently recorded for training purposes!) And voila, he is all set to fake your voice!

Such scams are on the rise worldwide. In banking circles, these scams, where fraudsters impersonate a known person or entity and convince people to transfer money to them, are known as authorised push payment frauds or APP frauds. According to reports, it now accounts for 40% of UK bank fraud losses and could cost US\$4.6 billion in the US and the UK alone by 2026. According to data from the Federal Trade Commission (FTC), there were over 36,000 reports last year in the US of people being swindled by imposters. Of these, 5,100 happened over phone and accounted for over US\$11 million in losses. In Singapore too, the number of reported scam cases rose by 32.6% in 2022 compared to